Constructing g-computation estimators: two case studies in selection bias

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eAppendix

Appendix 1: Standard g-computation

A1.1: Identification Proof

Here, we prove the identification result provided in Equation (1). Note that we assume conditional treatment exchangeability (i.e., $Y^a \coprod A \mid X$) with positivity (i.e., $\Pr(A = a \mid X = x) > 0$ for all $a \in \{0,1\}, x \in \mathcal{X}$, where $\mathcal{X}$ is the support of X), conditional selection exchangeability (i.e., $Y^a \coprod S \mid A, X$) with positivity (i.e., $\Pr(S = 1 \mid A = a, X = x) > 0$ for all $a \in \{0,1\}, x \in \mathcal{X}$), and causal consistency (i.e., $Y_i^A = Y_i$). Therefore,

$$
\begin{aligned}
\mu^a &= E[Y^a] \\
&= E[E\{Y^a \mid X\}] \\
&= E[E\{Y^a \mid A = a, X\}] \\
&= E[E\{Y^a \mid A = a, X, S = 1\}] \\
&= E[E\{Y \mid A = a, X, S = 1\}]
\end{aligned}
$$

following from the definition of μ^a, the law of iterated expectation, conditional treatment exchangeability with positivity, conditional selection exchangeability with positivity, and causal consistency, respectively.

A1.2: Consistency and Asymptotic Normality

Recall that the estimating functions for standard g-computation are

$$
g_1(O_i; \hat{\theta}) = \begin{bmatrix} S_i\{Y_i - m(\mathbb{X}_i; \hat{\beta})\}\mathbb{X}_i^T \\ \hat{Y}^1 - \hat{\mu}^1 \\ \hat{Y}^0 - \hat{\mu}^0 \\ (\hat{\mu}^1 - \hat{\mu}^0) - \hat{\psi} \end{bmatrix}
$$

To begin, consider the first estimating function for a generic element $\mathbb{x}$ in the design matrix $\mathbb{X}$ without a loss of generality. To show that it is unbiased, note that

$$
\begin{aligned}
E[S\{Y - m(\mathbb{X}; \beta)\}\mathbb{x}] &= E[\{Y - m(\mathbb{X}; \beta)\}\mathbb{x} \mid S = 1] \Pr(S = 1) \\
&= E[\{Y\mathbb{x} - m(\mathbb{X}; \beta)\mathbb{x}\} \mid S = 1] \Pr(S = 1) \\
&= \{E[Y\mathbb{x} \mid S = 1] - E[m(\mathbb{X}; \beta)\mathbb{x} \mid S = 1]\} \Pr(S = 1) \\
&= \{E[E(Y\mathbb{x} \mid A, X, S = 1) \mid S = 1] - E[m(\mathbb{X}; \beta)\mathbb{x} \mid S = 1]\} \Pr(S = 1) \\
&= \{E[\mathbb{x}\, E(Y \mid A, X, S = 1) \mid S = 1] - E[m(\mathbb{X}; \beta)\mathbb{x} \mid S = 1]\} \Pr(S = 1) \\
&= E[\mathbb{x}\, \{E(Y \mid A, X, S = 1) - m(\mathbb{X}; \beta)\} \mid S = 1] \Pr(S = 1) \\
&= E[\mathbb{x}\, \{E(Y \mid A, X, S = 1) - E(Y \mid A, X, S = 1; \beta)\} \mid S = 1] \Pr(S = 1) \\
&= E[\mathbb{x}\, \{E(Y \mid A, X, S = 1) - E(Y \mid A, X, S = 1)\} \mid S = 1] \Pr(S = 1) \\
&= 0
\end{aligned}
$$

following from 1) law of total expectation, 2) distribution of $\mathbb{x}$, 3) sum of expectations, 4) law of iterated expectation, 5) $\mathbb{x}$ must either be A, X, or some transformation of A, X, 6) sum of expectations, 7) definition of $m(\mathbb{X}; \beta)$, 8) correct model specification, and 9) identity. As this was proven for a generic element in the design matrix, this proof of unbiasedness suffices to show the estimating equation is unbiased for all elements in $\mathbb{X}$. Additionally, if one adopts a perspective of regression as an orthogonal projection, then correct model specification is not necessary as an assumption here. As the following proofs require correct model specification, we assume so here as well to simplify exposition. Now consider the second and third estimating functions. For ease, we prove

$$
\begin{aligned}
E\left[\hat{Y}^a - \mu^a\right] &= E\left[\hat{Y}^a\right] - \mu^a \\
&= E[E\{Y \mid A = a, X, S = 1\}] - \mu^a \\
&= E[Y^a] - \mu^a \\
&= E[Y^a] - E[Y^a] = 0
\end{aligned}
$$

for $a \in \{0,1\}$. Here, 1) follows from the expectation of a constant, 2) follows from the definition of $\hat{Y}^a$ and correct model specification, 3) follows from the identification proof, and 5) follows from the definition of μ^a. Finally, it immediately follows that $E[(\mu^1 - \mu^0) - \psi] = 0$ from the definitions of ψ and μ^a.

Having shown each of the corresponding estimating equations to be unbiased at θ, it follows under suitable regularity conditions that $\sqrt{n}\left(\hat{\theta} - \theta\right) \rightarrow^d N\left(0, \mathbb{V}(\theta)\right)$ (i.e., the difference between our estimator and the truth converges to a normal distribution with variance $\mathbb{V}(\theta)$ at a rate of $\sqrt{n}$), where $\mathbb{V}(\theta) = \mathbb{B}(\theta)^{-1}\mathbb{M}(\theta)[\mathbb{B}(\theta)^{-1}]^T$ is the sandwich variance with $\mathbb{B}(\theta) = E[-\partial g(\theta)/\partial\theta]$ is the expectation of the gradient of the estimating functions and $\mathbb{M}(\theta) = E[g(\theta)g(\theta)^T]$ is the expectation of the outer product of the estimating functions.[16]

Appendix 2: G-computation for Case 1

A2.1: Identification Proof

Here, we prove the identification result provided in Equation (2). As implied by Figure 1, we now assume marginal treatment exchangeability (i.e., $Y^a \coprod A$) with positivity (i.e., $\Pr(A = a) > 0$ for all $a \in \{0,1\}$). For conditional selection exchangeability, we assume that $Y^a \coprod S \mid A = a, X^a$, as implied by Figure 1. This comes paired with the positivity assumption that $\Pr(S = 1 \mid A = a, X^a = x) > 0$ for all $a \in \{0,1\}, x \in \mathcal{X}^a$. Finally, we assume causal consistency (i.e., $X^A = X$, $S^A = S$ and $Y^A = Y$).[26] Therefore,

$$\begin{aligned}\mu^a = E[Y^a] &= E[Y^a \mid A = a] \\ &= E[E(Y^a \mid A = a, X^a) \mid A = a] \\ &= E[E(Y^a \mid A = a, X^a, S^a = 1) \mid A = a] \\ &= E[E(Y \mid A = a, X, S = 1) \mid A = a]\end{aligned}$$

following from marginal treatment exchangeability with positivity, law of iterated expectation, conditional selection exchangeability with positivity, and causal consistency, respectively.

A2.2: Consistency and Asymptotic Normality

Recall that the proposed g-computation estimating functions are

$$g_2(O_i; \hat{\theta}) = \begin{bmatrix} S_i\{Y_i - m(\mathbb{X}_i; \hat{\beta})\}\mathbb{X}_i^T \\ A_i(\hat{Y}_i^1 - \hat{\mu}^1) \\ \{1 - A_i\}(\hat{Y}_i^0 - \hat{\mu}^0) \\ (\hat{\mu}^1 - \hat{\mu}^0) - \hat{\psi} \end{bmatrix}$$

It follows that the first estimating equations is unbiased following the same proof as in Appendix 1.2. Now, consider the second estimating function for $a \in \{0,1\}$,

$$\begin{aligned}E[I(A = a)(\hat{Y}^a - \mu^a)] &= E[\hat{Y}^a \mid A = a] \Pr(A = a) - E[\mu^a \mid A = a] \Pr(A = a) \\ &= E[\hat{Y}^a \mid A = a] \Pr(A = a) - \mu^a \Pr(A = a) \\ &= E[E(Y \mid A = a, X) \mid A = a] \Pr(A = a) - \mu^a \Pr(A = a) \\ &= E[Y^a] \Pr(A = a) - \mu^a \Pr(A = a) \\ &= E[Y^a] \Pr(A = a) - E[Y^a] \Pr(A = a) = 0\end{aligned}$$

following from 1) sum of expectations and law of total exchangeability, 2) μ^a being a constant, 3) definition of $\hat{Y}^a$ under correct model specification, 4) follows from the identification proof, and 5) follows from definition of μ^a. Therefore, both the second and third estimating equations are unbiased. Again, the final estimating equation is unbiased following the same proof as the one provided in Appendix 1.2. Therefore, it follows that $\sqrt{n}(\hat{\theta} - \theta) \to^d N(0, \mathbb{V}(\theta))$.

A2.3: Data Generating Mechanism

The following data generating mechanism was used:

$$A \sim \text{Bernoulli}(0.5)$$

$$U \sim \text{Normal}(\mu = 0, \sigma = 1)$$

$$X \sim \text{Normal}(\mu = -1 + 2A + U, \sigma = 1)$$

$$S \sim \text{Bernoulli}\big(\text{expit}(2 - 1X)\big)$$

$$Y \sim \begin{cases} \text{Bernoulli}\big(\text{expit}(0.5 + 0.75U - 1A)\big) & \text{if } S = 1 \\ -9999 & \text{if } S = 0 \end{cases}$$

The true value of ψ (as approximated by simulating 10 million observations with S set to 1 for all observations) was -0.219. In the observed data, approximately 20% of participants had missing values of Y.

A2.4: Inverse Probability Weighting Estimator

The corresponding IPW expression for the identification results in A2.1 is

$$\mu^a = E\left[Y \times I(A = a) \times \frac{S}{\Pr(S = 1 \mid X)}\right]$$

and the corresponding estimating functions for the Hajek IPW are

$$\begin{bmatrix} \{S_i - \text{expit}(\mathbb{S}_i; \hat{\alpha})\}\mathbb{S}_i^T \\ \dfrac{S_i}{\text{expit}(\mathbb{S}_i; \hat{\alpha})} A_i(Y_i - \hat{\mu}^1) \\ \dfrac{S_i}{\text{expit}(\mathbb{S}_i; \hat{\alpha})} \{1 - A_i\}(Y_i - \hat{\mu}^0) \\ (\hat{\mu}^1 - \hat{\mu}^0) - \hat{\psi} \end{bmatrix}$$

where $\text{expit}(\mathbb{S}_i; \hat{\alpha})$ is a logistic regression model for the probability of S given X (i.e., $\mathbb{S}$ is a design matrix that includes X, $\text{expit}(x) = \{1 + \exp(-x)\}^{-1}$). Note that we do not provide a proof of the asymptotic properties of this estimator as it is beyond the scope of this paper.

Appendix 3: G-computation for Case 2

A2.1: Identification Proof

As in the main paper, notation is recycled from the prior case. Here, we prove the identification result provided in Equation (3). As implied by Figure 2, we have the following exchangeability assumptions (with their corresponding positivity assumptions): $Y^a \coprod A \mid Z$ and $Y^a \coprod S \mid A, Z, X$. Again, we assume causal consistency. Therefore,

$$\begin{aligned}
\mu^a = E[Y^a] &= E[E(Y^a \mid Z)] \\
&= E[E(Y^a \mid A = a, Z)] \\
&= E\{E[E(Y^a \mid A = a, Z, X) \mid A = a, Z]\} \\
&= E\{E[E(Y^a \mid A = a, Z, X, S = 1) \mid A = a, Z]\} \\
&= E\{E[E(Y \mid A = a, Z, X, S = 1) \mid A = a, Z]\}
\end{aligned}$$

following from the law of iterated expectations, conditional treatment exchangeability with positivity, the law of iterated expectations, conditional selection exchangeability with positivity, and causal consistency, respectively.

A2.2: Consistency and Asymptotic Normality

Again, recall that the stacked estimating equations are

$$g_3(O_i; \hat{\theta}) = \begin{bmatrix} S_i\{Y_i - m(\mathbb{W}_i; \hat{\beta})\}\mathbb{W}_i^T \\ \{\hat{Y}_i^1 - m(\mathbb{V}_i; \hat{\gamma}_1)\}\mathbb{V}_i^T \\ \tilde{Y}^1 - \hat{\mu}^1 \\ \{\hat{Y}_i^0 - m(\mathbb{V}_i; \hat{\gamma}_0)\}\mathbb{V}_i^T \\ \tilde{Y}^0 - \hat{\mu}^0 \\ (\hat{\mu}^1 - \hat{\mu}^0) - \hat{\psi} \end{bmatrix}$$

A similar proof in Appendix 1.2 can be used to show $E[S\{Y - m(\mathbb{W}; \beta)\}\mathbb{w}] = 0$ for all $\mathbb{w} \in \mathbb{W}$ by substituting in (Z, X) in place of X. Now consider $E\left[\{\hat{Y}^a - m(\mathbb{V}; \gamma_a)\}\mathbb{v}\right]$ for $a \in \{0,1\}$,

$$\begin{aligned}
E\left[\{\hat{Y}^a - m(\mathbb{V}; \gamma_a)\}\mathbb{v}\right] &= E\left[\hat{Y}^a \mathbb{v}\right] - E[m(\mathbb{V}; \gamma_a)\mathbb{v}] \\
&= E\left[\hat{Y}^a \mathbb{v}\right] - E\left[E\{\hat{Y}^a \mid A, Z\}\mathbb{v}\right] \\
&= E\left[E\{\hat{Y}^a \mathbb{v} \mid A, Z\}\right] - E\left[E\{\hat{Y}^a \mid A, Z\}\mathbb{v}\right] \\
&= E\left[E\{\hat{Y}^a \mid A, Z\}\mathbb{v}\right] - E\left[E\{\hat{Y}^a \mid A, Z\}\mathbb{v}\right] = 0
\end{aligned}$$

where 1) follows from distribution of $\mathbb{v}$ and the sum of expectations, 2) follows from the definition of $m(\mathbb{V}; \gamma_a)$ under correct model specification, 3) follows from the law of iterated expectation, and 4) follows from $\mathbb{v}$ being A, Z, or some transformation of A, Z. Again, for $a \in \{0,1\}$,

$$\begin{aligned}
E[\tilde{Y}^a - \mu^a] &= E[\tilde{Y}^a] - \mu^a \\
&= E\left[E[E(Y \mid A = a, Z, X, S = 1) \mid A = a, Z]\right] - E[Y^a] \\
&= E[Y^a] - E[Y^a] = 0
\end{aligned}$$

following from the sum of expectations, the definitions of $\tilde{Y}^a$ and μ^a, and the identification proof. Again, it is apparent that $E[(\mu^1 - \mu^0) - \psi] = 0$ is unbiased as described in Appendix 1.2. Therefore, it follows that $\sqrt{n}(\hat{\theta} - \theta) \to^d N(0, \mathbb{V}(\theta))$.

A2.3: Data generating mechanism for simulation

The following data generating mechanism was used:

$$U_1 \sim \text{Bernoulli}(0.5)$$

$$U_2 \sim \text{Bernoulli}(0.5)$$

$$Z \sim \text{Bernoulli}(0.5)$$

$$A \sim \text{Bernoulli}\big(\text{expit}(-2.3 + \log(2)\, Z + \log(4)\, U_1)\big)$$

$$X \sim \text{Normal}(\mu = 4U_1 - 4U_2, \sigma = 1)$$

$$S \sim \text{Bernoulli}\big(\text{expit}(0.25X)\big)$$

$$Y \sim \begin{cases} \text{Bernoulli}\big(\text{expit}(-2 - 2A + \log(2)\, Z + \log(2)\, AZ + \log(4)\, U_2)\big) & \text{if } S = 1 \\ -9999 & \text{if } S = 0 \end{cases}$$

The true value of ψ (as approximated by simulating 10 million observations with S set to 1 for all observations) was -0.205. In the observed data, approximately 50% of participants had missing values of Y.

A3.4: Inverse Probability Weighting Estimator

The corresponding IPW expression for the identification results in A3.1 is

$$\mu^a = E\left[Y \times \frac{I(A = a)}{\Pr(A = a \mid Z)} \times \frac{S}{\Pr(S = 1 \mid X)}\right]$$

and the corresponding estimating functions for the Hajek IPW are

$$\begin{bmatrix} \{S_i - \text{expit}(\mathbb{S}_i; \hat{\alpha})\}\mathbb{S}_i^T \\ \{A_i - \text{expit}(\mathbb{A}_i; \hat{\alpha})\}\mathbb{A}_i^T \\ \dfrac{S_i}{\text{expit}(\mathbb{S}_i; \hat{\alpha})} \times \dfrac{A_i}{\text{expit}(\mathbb{A}_i; \hat{\alpha})} \times (Y_i - \hat{\mu}^1) \\ \dfrac{S_i}{\text{expit}(\mathbb{S}_i; \hat{\alpha})} \times \dfrac{1 - A_i}{1 - \text{expit}(\mathbb{A}_i; \hat{\alpha})} \times (Y_i - \hat{\mu}^0) \\ (\hat{\mu}^1 - \hat{\mu}^0) - \hat{\psi} \end{bmatrix}$$

where $\text{expit}(\mathbb{S}_i; \hat{\alpha})$ is a logistic regression model for the probability of S given X (i.e., $\mathbb{S}$ is a design matrix that includes X) and $\text{expit}(\mathbb{A}_i; \hat{\alpha})$ is a logistic regression model for the probability of A given Z (i.e., $\mathbb{A}$ is a design matrix that includes Z). Note that we do not provide a proof of the asymptotic properties of this estimator as it is beyond the scope of this paper.